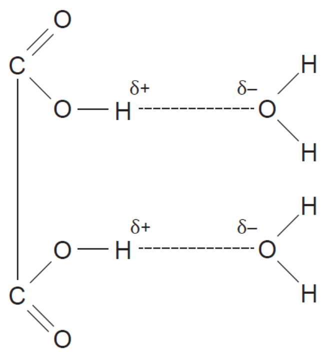
Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)		$\mathbf{2}\ C_4H_9OH\ +\ \mathbf{2}\ CO\ +\ \mathbf{½}\ O_2$		1		1		
		(ii)		award (1) for any of following cost of the catalyst separation of liquid materials from the catalyst cost of (maintaining) very high pressure accept any other valid suggestion		1		1		
		(iii)		ethanedioic acid is able to **hydrogen bond** to water molecules (hence its solubility) (1)  (1) hydrogen bonding to just one water molecule is acceptable the ester does not contain an O—H bond, so cannot hydrogen bond with water (hence largely insoluble) (1)	1 1	1		3		

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		moles of $(COO)_2Ca$ present = $\frac{11.52}{128}$ = 0.0900 (1) 1:1 ratio therefore 0.0900 mol of H_2SO_4 needed volume needed = $\frac{0.0900}{2} \times 1000$ = 45.0 cm^3 (1)	1	1		2	1	
		(ii)		7.15g of $(COOH)_2.2H_2O$ dissolves in 50 cm^3 of solution at 20ºC (1) total moles of $(COOH)_2.2H_2O$ present is 0.0900 total mass of $(COOH)_2.2H_2O$ formed = 0.0900 × 126 = 11.34 g (1) therefore, mass crystallising out = 11.34 – 7.15 = 4.19 g (1) ecf possible from incorrect moles in (b)(i)	1	1	1	3	2	
	(c)			$3CF_2H_2 + 3(COO)_2Na_2 \rightarrow C_6F_6 + 3Na_2CO_3 + 3H_2O$ award (1) for all formulae correct award (1) balancing only if all formulae correct		2		2		

Question				Marking details	Marks available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
	(d)			percentage of mass lost from graph = $\frac{5.40 - 4.73}{5.40} \times 100$ = 12.4% (1) percentage 'mass lost' in M_r = $\frac{28}{225} \times 100$ = 12.4% equation must be correct as the values are the same (1) accept same reasoning using 'remaining mass' ⇨ 87.6% *alternative method* moles of barium ethanedioate = $\frac{5.4}{225}$ = 0.024 (1) moles of barium carbonate = $\frac{4.73}{197}$ = 0.024 same number of moles therefore 1:1 stoichiometry (1)		1	1	2	1	
	(e)			ethylbenzene		1		1		
				Question 9 total	**4**	**9**	**2**	**15**	**4**	**0**